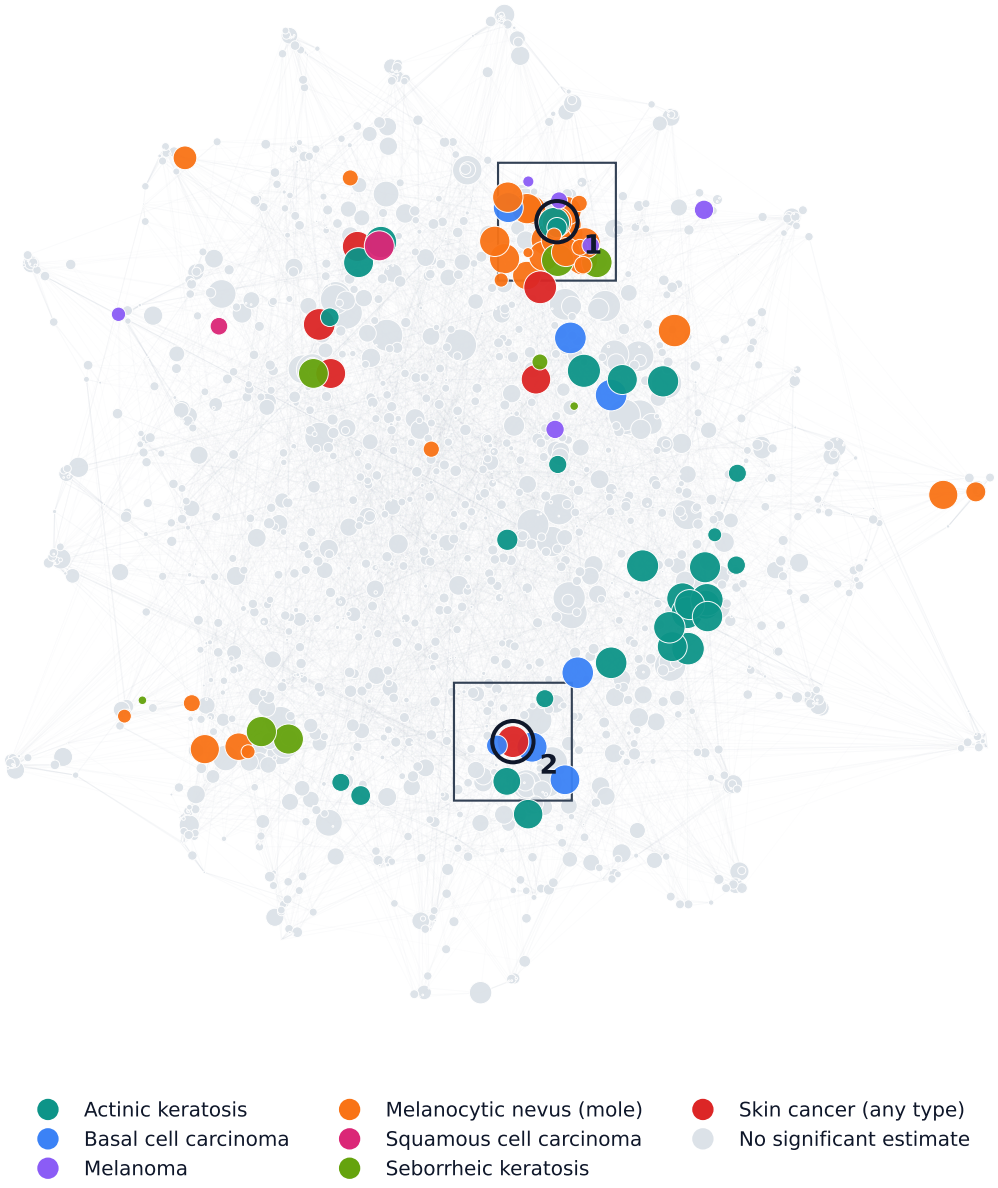
GLYPH / DISCOVERY BRIEF

Glyph + DINOv3: smartphone skin-lesion diagnosis

164 significant associations · 1600 concepts mapped

17/138 evaluable hits reproduce in test; 26 discovery hits lack evaluable comparison estimates.

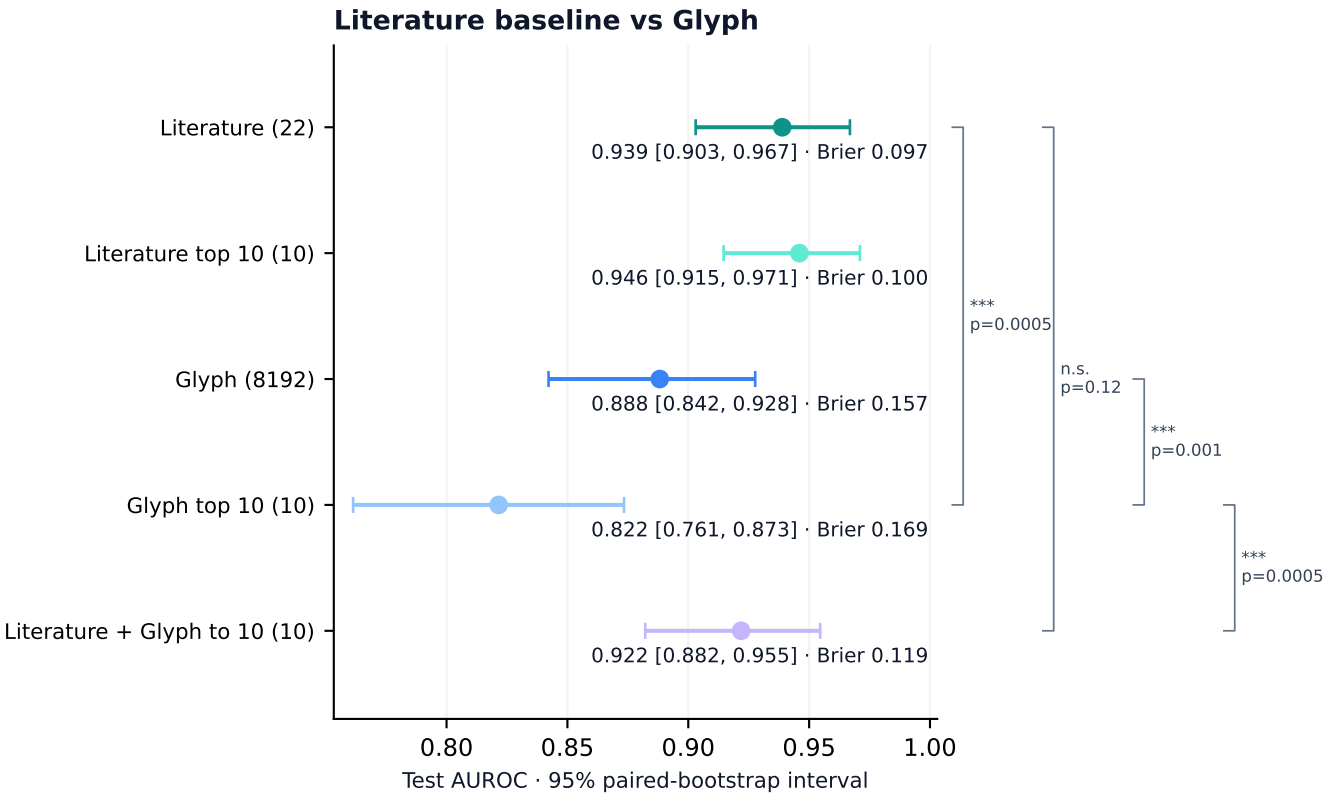
train 961 / val 206 / test 206
Reconstruction R^2: validation 0.757; test 0.759 · k=80 · below 0.85 target
Test probe AUC: input vectors 0.893; full dictionary 0.884



PAD-UFES-20: 1,373 patients, one original photograph each, seven diagnostic outcomes. Color: strongest significant outcome. Node area is proportional to activation prevalence, with a tiny visibility floor and a common density adjustment. Missing prevalences use a uniform small marker. Lines show saved correlations, with at most eight per concept.

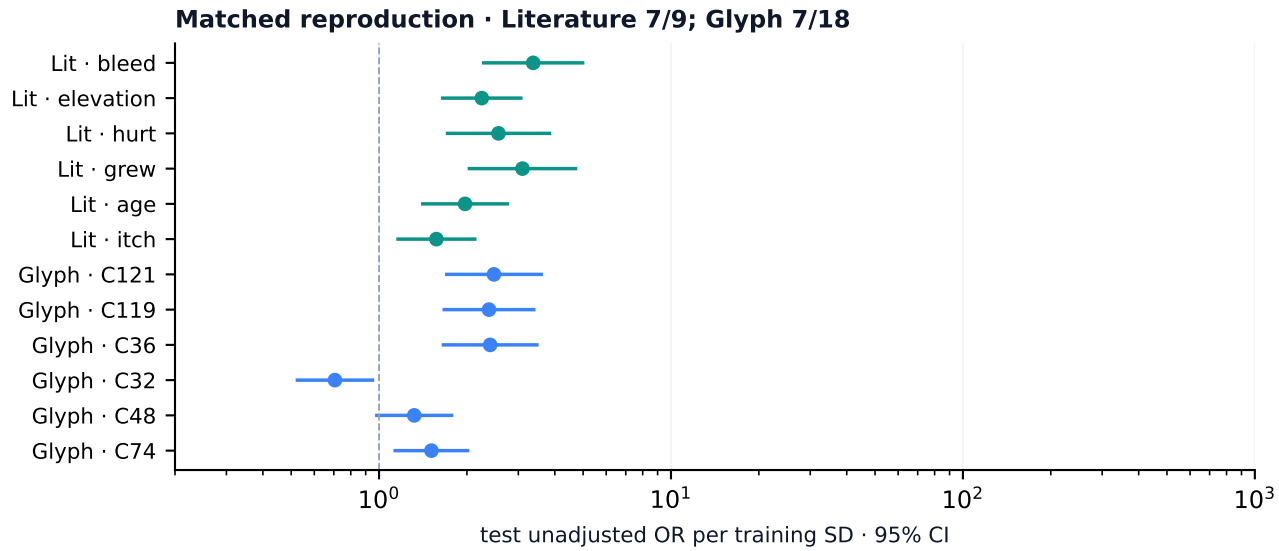
Patient-held-out diagnostic benchmark, not prognosis or external validation. The clinical baseline refits eight published inputs; it is not the historical fitted classifier. Some non-cancer labels use clinical consensus. Melanoma has only five test examples. Automated interpretation and confounder adjustment were not run. Images: Pacheco et al., PAD-UFES-20 v1, doi:10.17632/zr7vgbcyr2.1, CC BY 4.0; resized for display. Automated interpretation was not run.

PREDICTION / BASELINE COMPARISON



Capped combination: ΔAUC -0.0170 (95% CI -0.0388 to +0.0044), paired p=0.122 (exploratory, unadjusted).

Feature rankings use training only; validation selects ridge penalties and the best optimizer checkpoint. Combined models penalize literature and Glyph separately. Parentheses give predictor counts. The capped combined model pools literature and Glyph candidates and selects 10 total predictors using their training-only ranking; either or both sources may be selected.

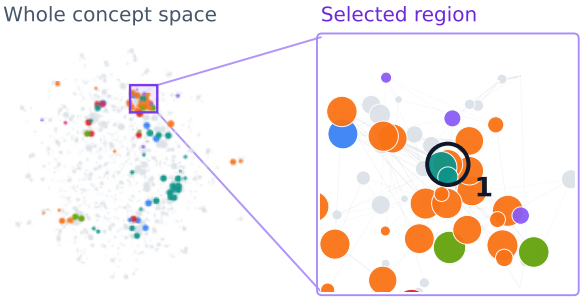


22 available literature markers; 8192 available Glyph concepts. Individual associations use 22 per source (44-test correction), separately from the full discovery family of 5642. Forest: up to six training-ranked markers per source; counts use the full matched set. Brackets show paired-bootstrap p-values, unadjusted (*<.05, **<.01, ***<.001; n.s. otherwise). Bootstrap intervals condition on fitted models. These results do not establish novelty or external validation.

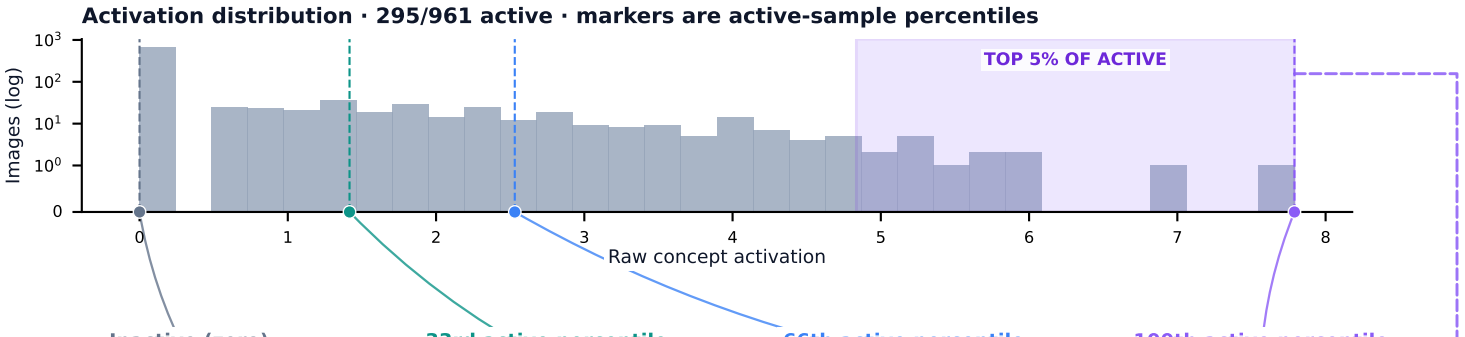
Literature source: Pacheco and Krohling, The impact of patient clinical information on automated skin cancer [...]

GLYPH / CONCEPT EVIDENCE

image_path concept 43 → Melanocytic nevus (mole)



train: OR 3.19 (95% CI 2.64–3.84); corrected p=2.9e-30
test: OR 3.95 (95% CI 2.50–6.25); corrected p=2.3e-05



Inactive (zero)



PAT_1000_31_620 · a=0
train · Other diagnosis

33rd active percentile



PAT_1904_3794_215 · a=1.42
train · Melanocytic nevus (mole)

66th active percentile



PAT_1451_1560_616 · a=2.53
train · Melanocytic nevus (mole)

100th active percentile

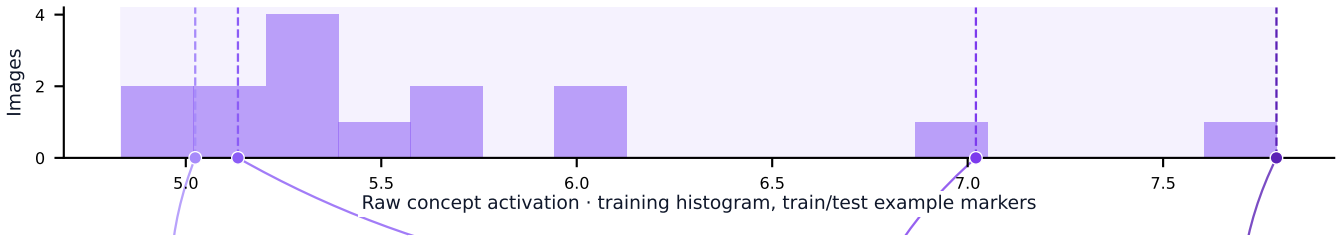


PAT_1516_1765_530 · a=7.79
train · Melanocytic nevus (mole)

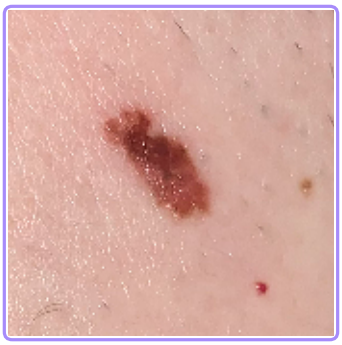
95–100% magnified

MAGNIFIED HIGH-ACTIVATION TAIL

Training-active 95th–100th percentiles shaded; strong train/test examples marked

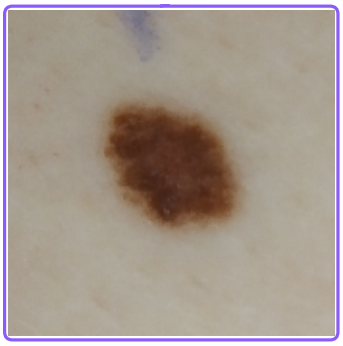


test · high activation



PAT_898_1706_496 · a=5.02
test · Melanocytic nevus (mole)

test · high activation



PAT_627_1188_503 · a=5.13
test · Melanocytic nevus (mole)

train · high activation



PAT_1257_887_828 · a=7.02
train · Melanocytic nevus (mole)

train · high activation

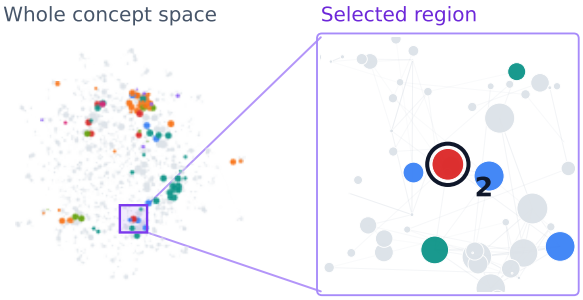


PAT_1516_1765_530 · a=7.79
train · Melanocytic nevus (mole)

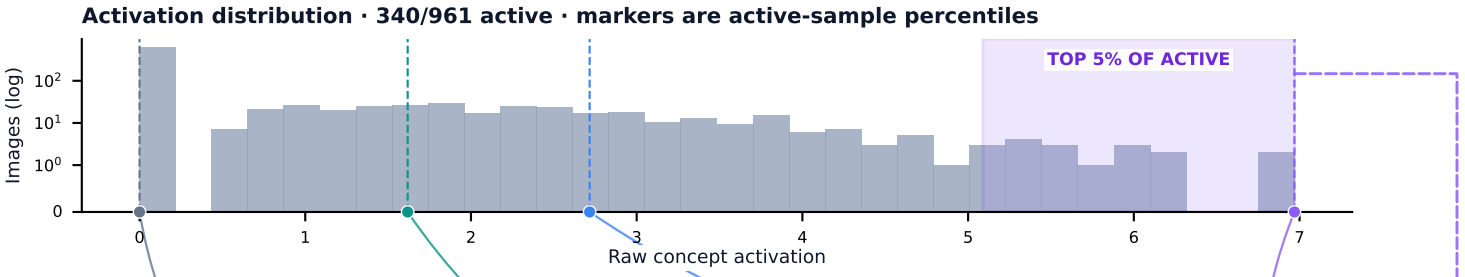
Scale images selected on training activations only. Zero is shown separately; other percentiles exclude zeros. Strong examples are selected by activation within train/test, without outcome-based selection. Images do not validate a semantic interpretation.

GLYPH / CONCEPT EVIDENCE

image_path concept 121 → Skin cancer (any type)



train: OR 2.81 (95% CI 2.34–3.36); corrected p=1.4e-25
test: OR 2.48 (95% CI 1.71–3.59); corrected p=0.011





PAT_1000_31_620 · a=0
train · No skin cancer

PAT_697_1317_892 · a=1.62
train · Skin cancer

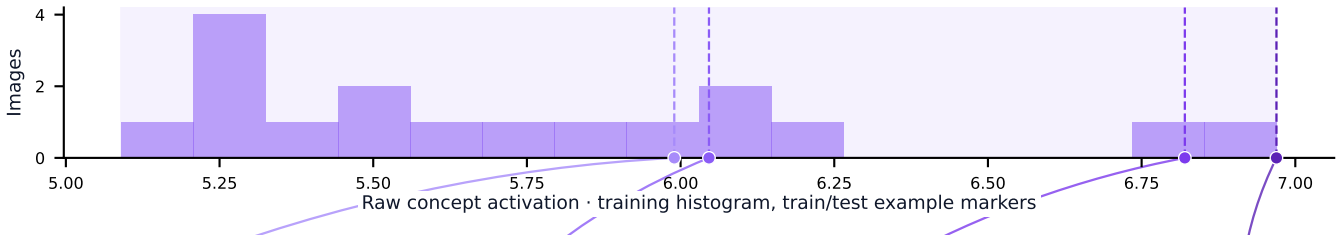
PAT_35_48_43 · a=2.72
train · Skin cancer

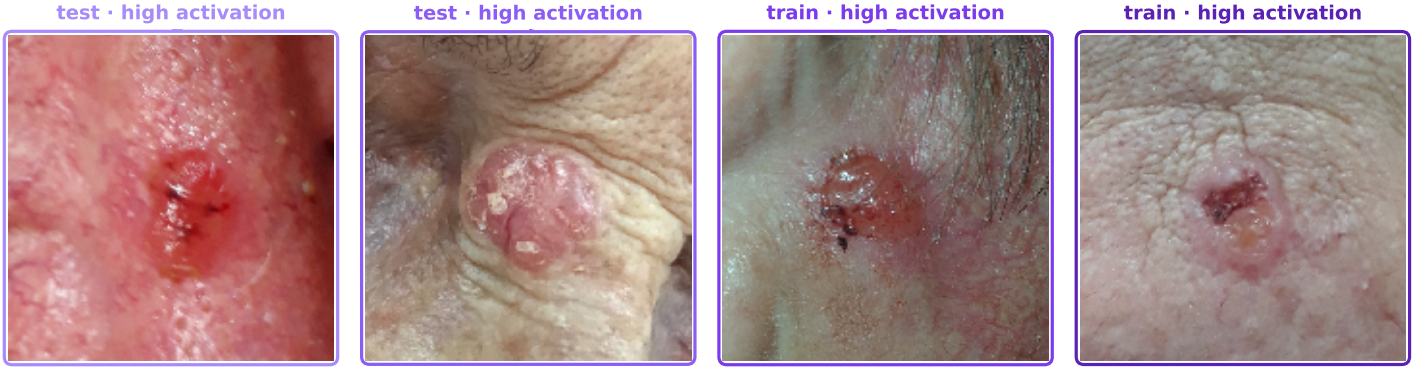
PAT_119_181_684 · a=6.97
train · Skin cancer

95–100% magnified

MAGNIFIED HIGH-ACTIVATION TAIL

Training-active 95th–100th percentiles shaded; strong train/test examples marked





PAT_229_349_141 · a=5.99
test · Skin cancer

PAT_785_1479_108 · a=6.05
test · Skin cancer

PAT_840_1595_981 · a=6.82
train · Skin cancer

PAT_119_181_684 · a=6.97
train · Skin cancer

Scale images selected on training activations only. Zero is shown separately; other percentiles exclude zeros. Strong examples are selected by activation within train/test, without outcome-based selection. Images do not validate a semantic interpretation.